

FINITE ELEMENT ANALYSIS (FEA) REPORT

Chaibasa Engineering College

Student: Rajnikant

Roll No: 259001004033

Branch: Mechanical Engineering (2nd Semester)

Subject: Mechanical Design

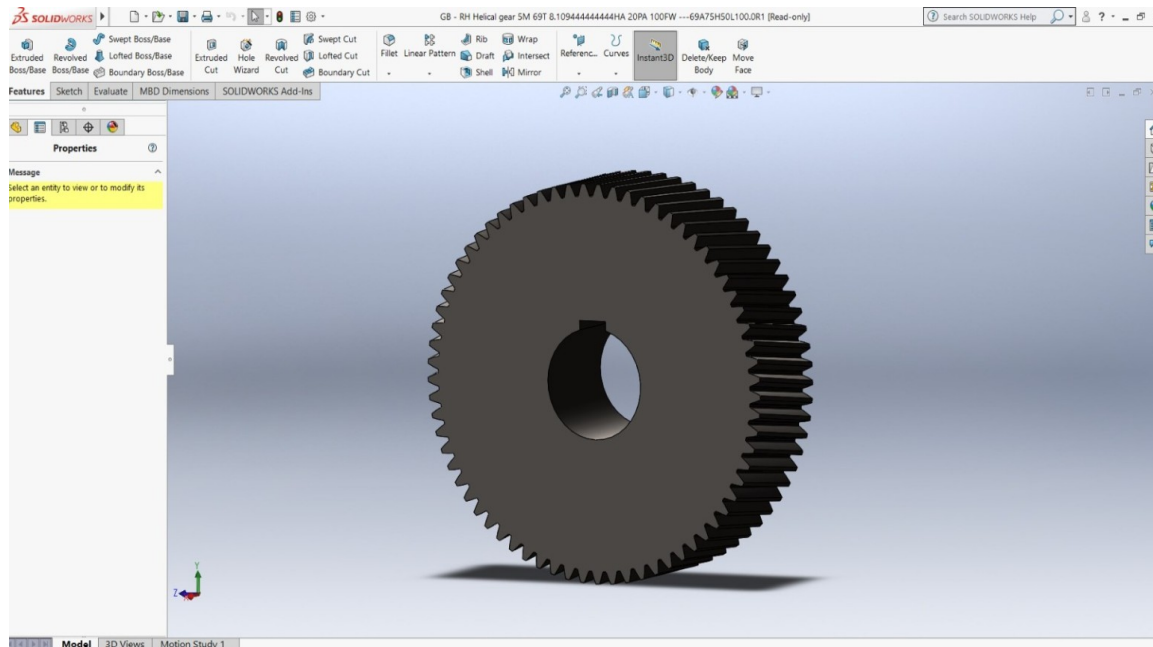
Faculty: Deepak Kumar

1. Introduction

Finite Element Analysis (FEA) is a numerical technique used to predict stresses, deformation and safety of engineering components before manufacturing.

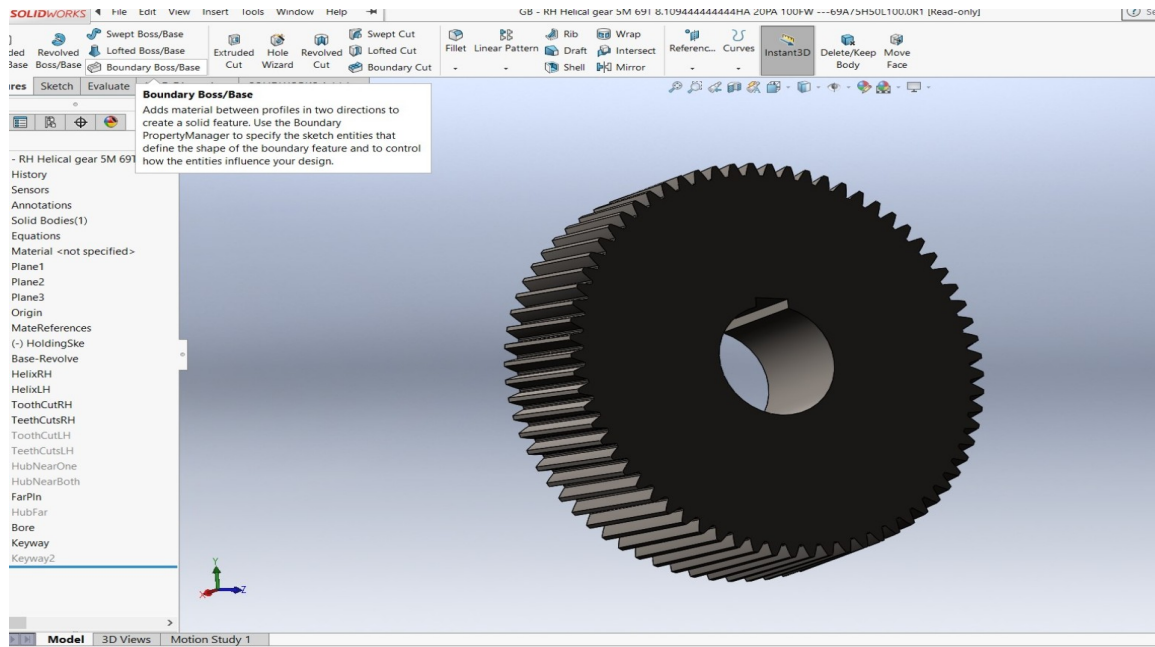
2. CAD Model

3D bracket model prepared in CAD software for structural analysis.



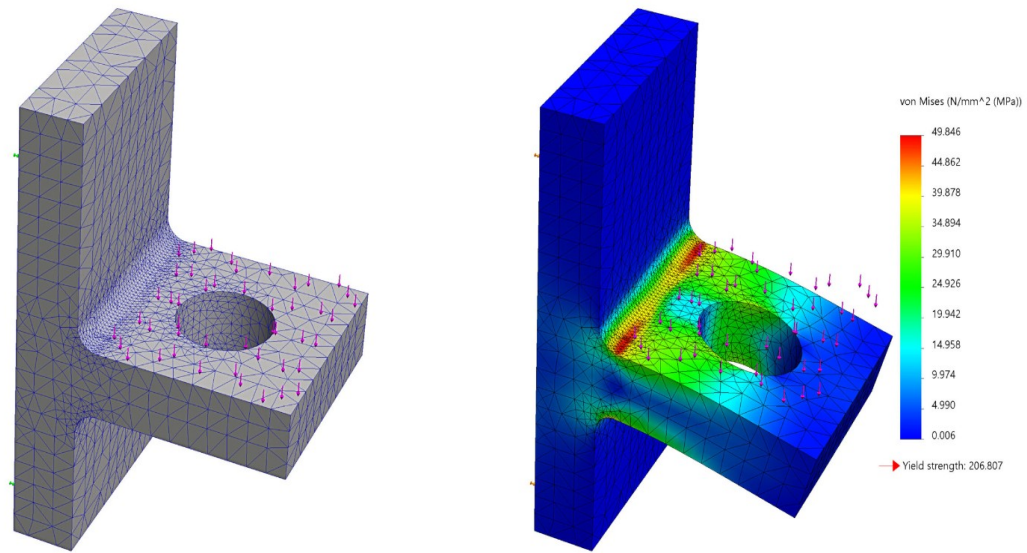
3. Geometry / Model View

Different views of the component used for simulation.



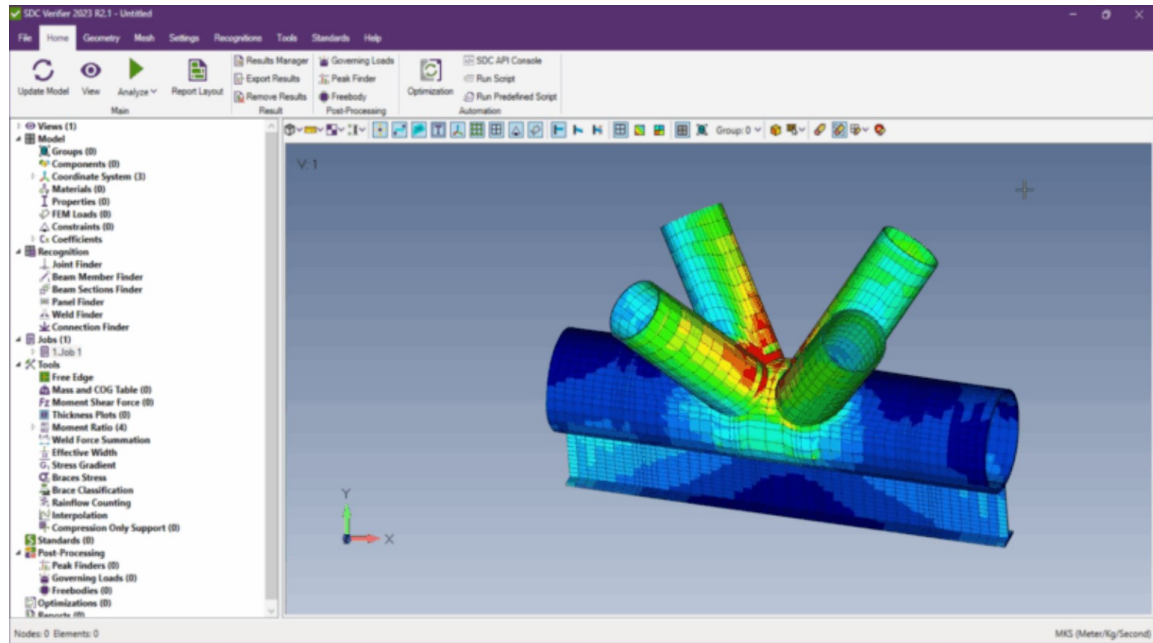
4. Mesh Generation

Tetrahedral mesh generated with finer elements around stress concentration regions.



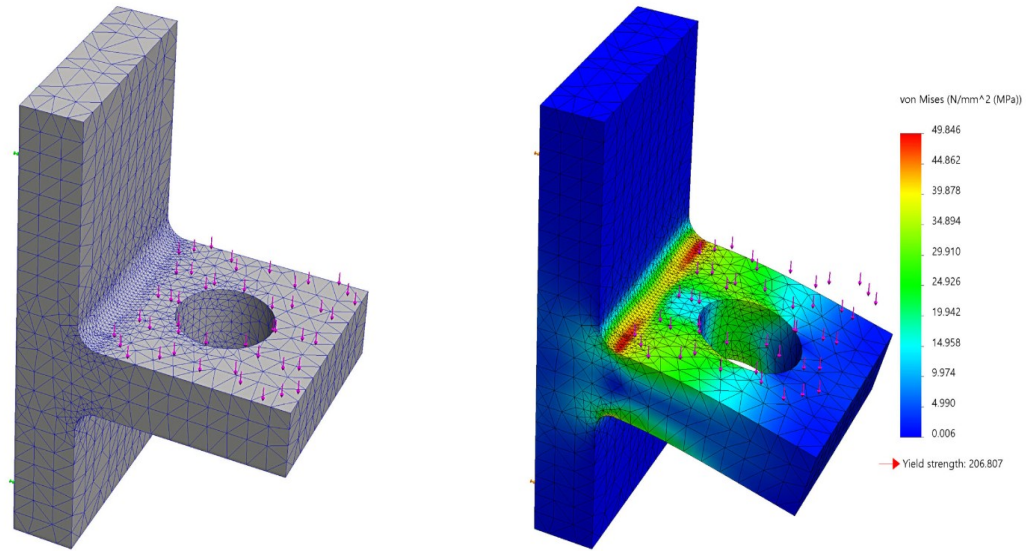
5. Boundary Conditions & Loading

One face fixed and external load applied on the horizontal arm.



6. Stress Distribution

Von Mises stress shows maximum stress near the fillet. Maximum stress remains below yield strength, indicating safe performance.



7. Results and Discussion

Observations:

- Maximum stress occurs at the fillet.
- Lowest stress appears away from the load.
- Maximum deformation occurs near the free end.
- Estimated Factor of Safety > 4 .

8. Conclusion

The FEA results indicate that the bracket is structurally safe under the applied loading. The design can be further optimized through mesh refinement and fatigue analysis.